Scalability

load parameters, fan-out and Amdahl's Law

Watch the 3-min video

In one line: scalability means handling more load by adding resources with proportional gains; it starts with picking the right load parameter, and it's limited by serial work and coordination.

1 What scalability really means

A system scales if adding resources gives roughly proportional capacity: double the servers, roughly double the load handled. If doubling servers gives 10% more, it doesn't really scale, however many servers you can afford.

2 First, define the load

“Load” depends on the system, and choosing the parameter that actually drives cost is the first step of any scaling discussion:

Load parameter	Example
Requests per second	An API at 20,000 RPS peak
Read/write ratio	A catalogue at 100 reads per write
Data size	5 TB growing 1 TB a year
Fan-out	One post delivered to N followers
Concurrent connections	1 M open WebSockets for chat

3 The Twitter example

Around 2012 Twitter handled about 4.6 thousand new tweets per second but about 300 thousand timeline reads per second. Building each timeline at read time was too expensive, so Twitter **pre-built timelines on write**: each tweet is copied into every follower's timeline.



Figure 1. Fan-out on write: expensive writes, cheap reads. The real load parameter is followers per tweet.

The average user has about 75 followers, but a celebrity has tens of millions, and one tweet would trigger millions of writes. So celebrity tweets are **fetched at read time** and merged in. The lesson: find the

parameter that actually hurts (here fan-out, not tweets per second) and design around it.

4 Why scaling is rarely linear

Real systems scale **sublinearly**: the curve bends and flattens because of coordination between machines, shared locks and contention, network overhead, and work that can't run in parallel. Amdahl's Law puts a number on the last one:

```
speedup(N) = 1 / ( s + (1 − s) / N )   ·   s = serial fraction
```

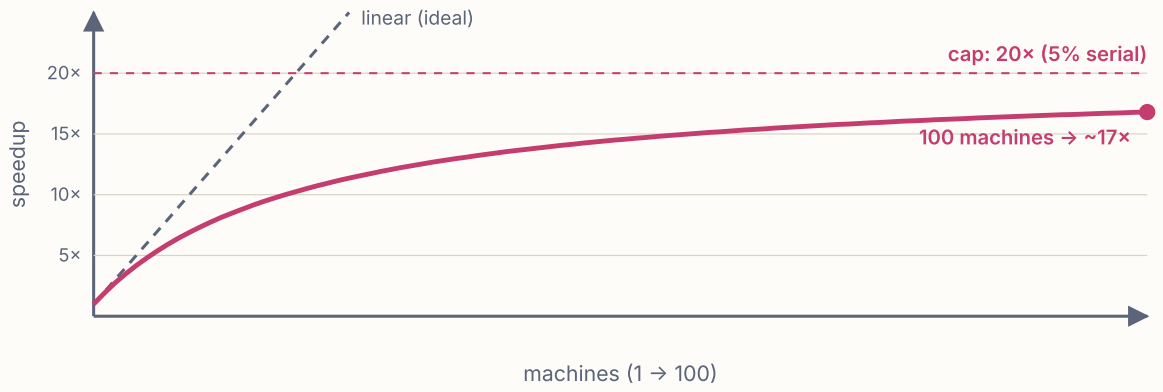


Figure 2. With 5% serial work, speedup can never exceed 20×; 100 machines give only about 17×.

WHAT TO DO ABOUT IT

Shrink the serial part: avoid global locks, partition data so machines don't coordinate, make work independent per user or per key, and move coordination off the hot path.

COMMON MISTAKES

- Saying “scalable” without naming the load parameter.
- Designing for RPS when fan-out or data size is the real driver.
- Expecting linear speedup from more machines.
- One global counter or lock that every request touches.

5 Key terms

Scalability: more resources → proportionally more capacity

Load parameter: the number that drives cost

Fan-out: work one action triggers

Sublinear: gains shrink as you add machines

Amdahl's Law: serial work caps speedup

Contention: work waiting on a shared resource

6 Interview questions

Q1. How do you define load for a system? BASIC

Pick the parameters that drive cost: requests per second, read/write ratio, data size, fan-out, concurrent connections. The right one depends on the system.

Q2. What limits linear scaling? BASIC

Serial work, coordination, contention and network overhead. Amdahl's Law: with a serial fraction s, speedup can't exceed 1/s.

Q3. **If 90% of the work is parallel, what's the maximum speedup?** BASIC
1 / 0.1 = 10×, no matter how many machines you add.

Q4. **Design a news feed for users who follow up to 5,000 accounts. Push or pull?** SCENARIO
Push (fan-out on write) for normal accounts so reads are cheap; pull at read time for accounts with huge follower counts; merge both when building the feed.

Q5. **You doubled the servers and throughput rose only 15%. What do you check?** SCENARIO
The shared bottleneck: the database, a global lock, a single-threaded component, a saturated network link, or a downstream rate limit. Profile and find what everyone waits on.

Q6. **What load parameter matters most for a chat app?** SCENARIO
Concurrent connections and message fan-out to group members, more than raw requests per second.

Q7. **Why did Twitter treat celebrities differently?** TRICKY
Fan-out on write for 30 million followers means 30 million writes per tweet, creating huge delays. Fetching celebrity tweets at read time avoids that spike.

Q8. **Is a system that handles 1 M RPS on 1,000 servers scalable?** TRICKY
Not necessarily. Scalability is about the slope: whether adding servers keeps adding capacity proportionally, and at what cost per request.

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